

like the fragment/nested layout.

Let's build it further such that the container has a column. Like any other widget, the column can be nested too. It stacks the items/widgets vertically. Let's say this column has two items: a row at the top and some text at the bottom. What is a row, you may ask? This is nothing but a widget — like I said before, everything is a widget. Row arranges the widgets horizontally. We can go deeper into the widget tree by adding some text and an icon in the row. What you need to remember is that when you want widgets to be positioned vertically (one on top of the other), use the column to lay them out, and when you want widgets to be positioned horizontally, use a row. To add text, use a text widget, and if you want to add an icon, use an icon widget. For images, use the image widget. Pretty simple, isn't it? Your widget tree will look something like what's shown in Figure 1 by the time you build the app.

Our widget tree is just a whole bunch of widgets that are nested within each other. Using these widgets, you can create a beautiful yet interactive UI. Now let's set up/download the tools to get started with app development.

Prerequisites for Flutter app development

Development environment setup:

Linux, Mac or Windows based computer

IDE: Android Studio/Visual Studio Code

Disk space: Minimum 10GB free space. Although Flutter requires only 1.64GB, you will need space to install Android Studio

Tools: Git

Note: For this article I will be covering only Android application development using a Windows system and Android Studio.

Before you begin, make sure you have a version of Java 11 installed and that the JAVA_HOME environment

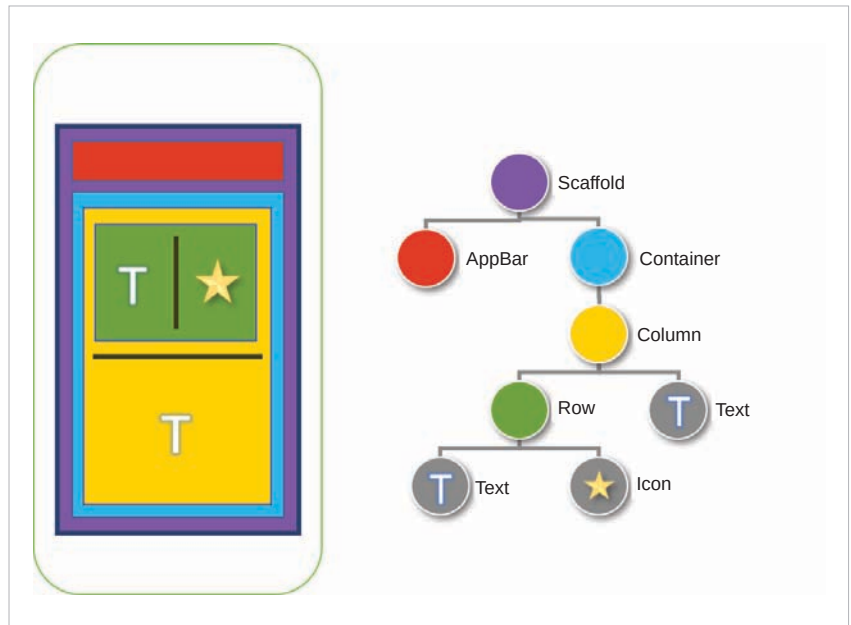


Figure 1: Anatomy of a Flutter app

variable is set to the JDK's folder. If you have installed Android Studio version 2.2 and higher, this should already be the case as it comes with a JDK.

There are three steps for completing the installation process:

- Install the Flutter SDK
- Install Android Studio
- Install Android Emulator

Installing the Flutter SDK

Download the Flutter SDK for Windows (<https://docs.flutter.dev/get-started/install/windows>) and follow the installation instructions. Once the installation is over, update the environment variable as mentioned in the installation guide. Let's verify that we have installed it correctly by running the following command:

```
Flutter -version
C:\Users\verma_ru>flutter --version
Flutter 3.7.6 • channel stable •
https://github.com/flutter/flutter.git
Framework revision 12cb4eb7a0 (7 days
ago) • 2023-03-01 10:29:26 -0800
Engine revision ada363ee93
Tools • Dart 2.19.3 DevTools 2.20.1
```

This will give you the installed version of Flutter and Dart. Now, run another command to check the environment details:

Flutter doctor

This will display a status report of the installation and the other software that we may need to install.

```
C:\Users\verma_ru>flutter doctor
Doctor summary (to see all details, run
flutter doctor -v):
[V] Flutter (Channel stable, 3.7.6,
on Microsoft Windows [Version
10.0.17763.557], locale en-IN)
[✓] Windows Version (Installed version
of Windows is version 10 or higher)
[✓] Android toolchain develop for
Android devices (Android SDK version
33.0.1)
[✓] Chrome - develop for the web
[✓] Visual Studio
develop for Windows (Visual Studio
Professional 2019 16.11.11)
[✓] Android Studio (version 2020.3)
[✓] Android Studio (version 2021.3)
[✓] Connected device (2 available)
[✓] HTTP Host Availability
• No issues found!
```